ĐẠI HỌC TÔN ĐỨC THẮNG
TON DUC THANG UNIVERSITY

502094 - Software Deployment, Operations And Maintenance

Course Introduction

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Course Overview

- Course code: **502094**
- Title: **Software Deployment, Operations and Maintenance**
- Credits: **3 (3,0)**
- Prerequisite:
  - Basic web development knowledge
  - Familiarity with command-line tools
  - Understanding of Git basics

Course Objective (Summary)

- **Core Skills**: Deploy, operate, maintain modern software systems
- **Technologies**: Linux, containers, cloud platforms, CI/CD, Infrastructure as Code
- **Focus**: Hands-on practice, team collaboration, measurable software delivery improvements

DevOps-Focused Nature of This Course

- Official title: **Software Deployment, Operations and Maintenance**
- Practical emphasis: **DevOps for web-based applications**
- We treat:
  - Deployment as **repeatable, automated pipelines**
  - Operations as **observability, reliability and scaling**
  - Maintenance as **continuous improvement and safe change management**
- Outcome:
  - You think like an engineer responsible for the **entire lifecycle** of a web service, not just its source code.

What “DevOps for Web Developers” Means Here

- You will learn to:
  - Take a **Node.js / Spring Boot** web application from laptop to **cloud production**
  - Design deployments that are **reproducible** and **scripted**, not one-off manual steps
  - Use Git, CI/CD and IaC so that **infrastructure and configuration are version-controlled**
- Emphasis on:
  - Skills that junior backend / DevOps engineers actually use in industry
  - Bridging the gap between **“I can code a feature”** and **“our system runs reliably in production”**

Course Outcomes — Highlight

1. **Linux, Shell & Git:** Manage systems, automate tasks with shell scripting, and maintain reproducible workflows using **Git**.
2. **Cloud Deployment (AWS):** Provision, configure and secure cloud resources; ensure **availability** and **performance**.
3. **Containers & Orchestration:** Build and operate **Docker** containers and **Kubernetes** clusters for scalable, portable applications.
4. **IaC & Observability:** Apply **Terraform**, configuration management, and monitoring to automate provisioning and gain **observability**.
5. **Collaboration & Professionalism:** Demonstrate teamwork, continuous learning, documentation and problem-solving in DevOps projects.

Brief Course Content

- **DevOps principles, lifecycle, and culture** for modern software delivery
- **Linux, Git & Shell scripting** fundamentals for automation
- **Secure server management** and **web deployment on AWS**
- **Containerization (Docker)** and **orchestration (Kubernetes)**
- **CI/CD** and **Infrastructure as Code (Terraform, Ansible)**
- **Monitoring, Security & Continuous Improvement** in production

Assessment & Grading

- **Process Evaluation:** `20%` — Lab completion, environment setup, and team process
- **Mid-term Test / Presentation:** `30%` — Team demo and design explanation
- **Final Examination / Report:** `50%` — Final project report demonstrating reproducible deployment and observability

Course Schedule (15 weeks)

- Module 1 — Introduction to DevOps & Course Setup
- Module 2 — Linux, Git & Shell Scripting Foundations
- Module 3 — Server Access, Remote Management & Tunneling
- Module 4 — Cloud Infrastructure with AWS
- Module 5 — Web Application Deployment Fundamentals
- Module 6 — Containerization with Docker
- Module 7 — Container Orchestration with Kubernetes
- Module 8 — Load Balancing & Application Scaling

Course Schedule (15 weeks)

- Module 9 — Continuous Integration (CI)
- Module 10 — Continuous Deployment (CD) & Release Management
- Module 11 — Infrastructure as Code (IaC)
- Module 12 — Configuration Management & Automation
- Module 13 — Monitoring, Logging & Observability
- Module 14 — DevSecOps & Pipeline Security
- Module 15 — Course Review & Retrospective

Recommended Primary Textbook

- Yevgeniy Brikman, **Fundamentals of DevOps and Software Delivery** (2025), O'Reilly — *Primary textbook for hands-on exercises*.

Supplementary & Additional Readings

- Mikael Krief, **Learning DevOps** (2022), Packt
- Bradley Smith, **DevOps for the Desperate** (2022), No Starch
- Mohamed Labouardy, **Pipeline as Code** (2021), Manning
- Rafał Leszko, **Continuous Delivery with Docker and Jenkins** (2022), Packt
- Yevgeniy Brikman, **Terraform: Up and Running** (2022), O'Reilly
- Justin Domingus & John Arundel, **Cloud Native DevOps with Kubernetes** (2022), O'Reilly

Important Notes for a Hands-on Course

- Emphasis on **practical labs**: you will *build, run and break* things to learn how to fix them.
- Use **GitHub** for versioning; preferably a single team repo per project.
- Prefer local reproducible environments: **Docker** or **VMs** / **WSL2** for Windows users.
- Keep all infrastructure and deployment steps as **Infrastructure as Code (IaC)**.

Starter tips & expectations (for students)

- Install: **Docker Desktop**, **VS Code**, **Git**, and a Linux environment (WSL2 / VM / cloud)
- Work in small increments: **commit early, commit often**
- Write clear **README.md** and runbook for your project
- Practice **blameless retrospectives** after each sprint/lab

Quick contact & office hours

- Instructor: **MSc. Mai Van Manh**
- Email: **maivanmanh@tdtu.edu.vn**
- Office hours: announced in LMS or by appointment

Closing

- Understand the course goals and assessment structure
- Prepare a working development environment for the semester
- Clone the starter repo and run the sample lab
- Know where to find learning resources and support